

Images are data

A digital image is a rectangular collection of measurements. Shape, channel order, value range, and coordinates form a contract.

15–20 MINUTES

Prerequisite: Lesson 01

INTERACTIVE SLIDER

1 • Pixel grid

Rows and columns of samples.

2 • Channels

Several measurements per pixel.

3 • Tensor

Shape and numeric type.

4 • Transform

Resize and normalize consistently.

Pixels and coordinates

A color image can be represented as a three-dimensional array: height $\times$ width $\times$ channels. A 480-row, 640-column RGB image commonly has shape `(480, 640, 3)`. Pixel coordinates usually start at the top-left: x increases to the right and y increases downward.

A pixel is a sample over a small image area. Its three channel values do not mean “red object” or “cube.” They record intensity in color channels. Illumination, exposure, sensor processing, surface reflectance, and shadows all influence those numbers.

Central mental model

An image tensor is meaningful only with its contract: axis order + channel order + value range + coordinate transform.

RGB, BGR, HWC, and CHW

RGB names channels red, green, blue. OpenCV commonly stores color images in BGR order. Many neural models accept channels-first tensors, written `CHW`, while camera or

image libraries commonly use `HWC`. Rearranging axes changes where values live; it does not change the scene.



Value ranges and normalization

An 8-bit channel stores integers from 0 to 255. A model may expect floating-point values from 0 to 1, produced by dividing by 255. This is not cosmetic: a model trained on one representation may behave badly if inference uses another.

Interactive normalization

8-bit channel value: **128**



Normalized value: **0.502**

Move the slider. Notice that normalization changes the numeric scale while preserving ordering: brighter channel samples remain larger.

Resize versus letterbox

Models often require a fixed input size. Stretching a rectangular image directly to a square changes shapes. **Letterboxing** scales the image uniformly and pads the unused region, preserving aspect ratio. A box predicted on the padded canvas must later be mapped back to the original coordinates.

This gives a general rule: every geometric transform applied before inference needs a corresponding mapping for spatial outputs.

Applied to our cubes

REPOSITORY OBSERVATION The current node receives OpenCV BGR data, converts it to RGB, letterboxes it to 640×640 , transposes HWC to CHW, converts to `float32`, divides by 255, and adds a batch dimension. The documented model input is `[1, 3, 640, 640]`.

GENERAL PRINCIPLE The exact size 640 and the axis order are artifact-specific. The need to keep preprocessing consistent with training applies broadly.

Project anchors: detection node and model contract.

Check your understanding

What does HWC → CHW change?

- ☐ The observed scene itself
- ☐ The camera exposure setting
- ☐ The array axis order

Explain it back

Describe the full contract of an image tensor. Then explain why a correct model can still fail when its channel order or numeric range is wrong.

Primary sources

- Szeliski — image formation and digital image representations.
- OpenCV geometric transforms — resizing, interpolation, and borders.